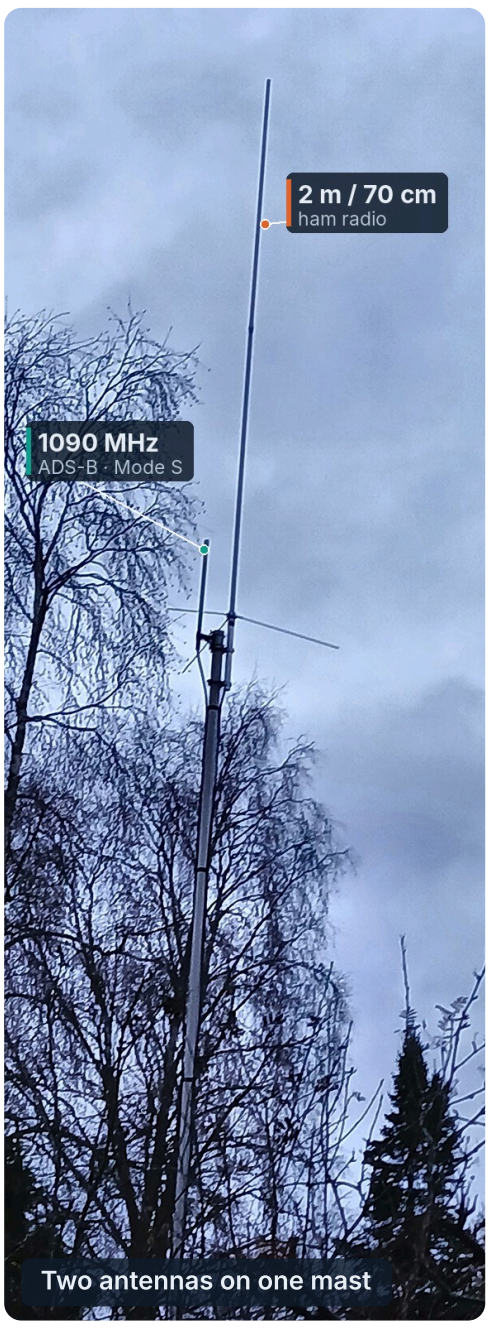
MODE-S Wind · the receiving station

Airliners as weather sensors — wind, temperature and the heights of the pressure surfaces from the Mode S and ADS-B replies heard in Tuusula, north of Helsinki-Vantaa (EFHK)

Otso Laakso · OH2GAX
Version v.08112026
November 2026 · 1 / 10

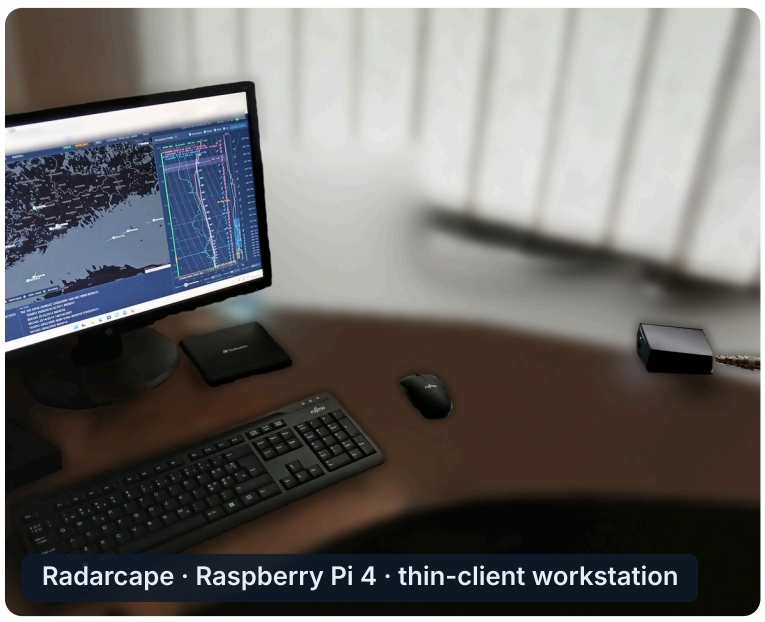
1 HARDWARE — one antenna, one receiver, one small computer (more receivers: pages 9–10); the users are served from the cloud (page 4)



Two antennas on one mast



Jetvision Radarcape EVO


Radarcape · Raspberry Pi 4 · thin-client workstation

Signal chain

Aircraft transponder › **1090 MHz** › GP antenna › coax › **Radarcape** › Ethernet › **Raspberry Pi 4** › browser · phone

Antenna
Two antennas share the mast: the **short ground-plane without radials** is the 1090 MHz ADS-B / Mode S antenna, the tall one with radials the author's 2 m / 70 cm ham-radio antenna. Aircraft at cruise levels are heard to about 160 NM.

Raspberry Pi 4
The local station: the Python collector with **pyModeS** decoding, the SQLite database, the hourly jobs and the web server — and the encrypted tunnel that feeds the cloud server.

Jetvision Radarcape EVO
An FPGA Mode S decoder with GPS-disciplined nanosecond timestamps — the basis of multilateration (MLAT). Every frame goes out as **Beast binary** over TCP (with its signal level), the aircraft list with MLAT positions as JSON.

Browser and phone
Viewers: the pages run in any browser, and `/app` gives the Skew-T, time–height, ILS approach, windshear, history and statistics on a phone (page 5).

2 WHAT THE AIRCRAFT SEND

The ground radars and Fintraffic's **wide-area multilateration (WAM)** network interrogate the aircraft on 1030 MHz, and the transponders answer on **1090 MHz**. The station only **listens**: it hears the answers, never the questions.

In Europe, Mode S **Enhanced Surveillance (EHS)** makes the larger aircraft answer with **Comm-B registers** — 56-bit snapshots of their own air-data and inertial systems: true airspeed, Mach, magnetic heading, roll, track and ground speed. Air traffic control uses them to see what the aircraft is doing; MODE-S Wind uses the same numbers to measure the air the aircraft flies through. On top come the aircraft's own **ADS-B** squitters with GNSS positions and the GNSS − barometric height difference.

Because the request is not heard, the reply does not say which register it holds. pyModeS infers it from the content (and finds the aircraft's address hidden in the reply's parity bits); ambiguous replies are dropped, and the Statistics page counts how often this happens.

≈ 130 /s
Mode S and ADS-B messages, 24-hour mean

1.3–1.9 s
median interval between one aircraft's BDS 5,0 replies (en route → final)

945 000
stored observations, 24 Sep – 26 Oct 2026

≈ 85 %
of the observations with a wind pair also give a temperature

FORMATS AND REGISTERS — used by MODE-S Wind

Format · register	Content
`DF17 ADS-B`	Sent by the aircraft itself: positions (TC 9–18), velocity with the **GNSS − baro height difference** (TC 19, about twice a second), operational status; NACp / NIC give the GNSS quality
`DF20 / DF21`	Comm-B reply to a radar or WAM request, with altitude (20) or squawk (21); carries one 56-bit register:
`BDS 5,0`	**Track and turn report**: roll angle, true track, ground speed, track-angle rate, true airspeed
`BDS 6,0`	**Heading and speed report**: magnetic heading, indicated airspeed, Mach, barometric and inertial vertical rate
`BDS 4,0`	Selected vertical intention: selected altitude and the pilot's QNH setting
`BDS 4,4 / 4,5`	Meteorological routine (MRAR) and hazard (MHR) reports — rare here; MHR temperatures are rejected (mostly other registers misidentified)
`BDS 1,0 · 1,7 · 2,0`	Data-link capability · supported registers · callsign
`DF11`	All-call reply — its parity carries the **interrogator code** (II / SI): which radar or WAM station is asking
`DF4 / DF5`	Surveillance replies: altitude, squawk — the reply rhythm per flight phase
`DF0 / DF16`	Airborne collision avoidance (ACAS / TCAS) traffic

Quality gates — GNSS values only with NACp ≥ 8 and outside the airframe's jamming episodes (from the GPS Quality monitor); MLAT keeps aircraft placed when jamming removes their ADS-B positions; WAM ground stations, helicopters and frozen or spoofed positions are filtered out before any meteorological use.

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3 FROM REPLIES TO WEATHER

Wind — ground vector minus air vector

```
W = (GS, true track) − (TAS, true heading)
true heading = magnetic heading
  + declination − δ(airframe)
```

A BDS 5,0 and a BDS 6,0 reply of the same aircraft at most 10 s apart; none while it banks or turns fast, and on final only from replies at most 6 s old. The declination comes from the World Magnetic Model 2025 at the aircraft's own position (8.7–12.6° around Helsinki) — 1° of heading is about 8 kt of wind at 450 kt.

Temperature — from the air-data computer

$$a = TAS / M, \quad T = a^2 / (\gamma R)$$
$$T\,[K] = 288.15 \cdot (TAS / (661.47\ kt \cdot M))^2$$

The static air temperature follows from the true airspeed (2 kt steps) and the Mach number (0.004 steps) of the same aircraft: about ±1.3 K per reply at cruise and ±3–4 K on approach, so profiles use trimmed means of many replies and aircraft.

Heights of the pressure surfaces

$$z = (\text{p-alt} + \Delta_{GNSS-baro}) \cdot 0.3048 - N_{EGM96} - \text{curve(alt)} - \text{offset}$$
$$T_v = g_0\ \Delta H / (R \ln(p_1/p_2))$$

The pressure altitude tells which pressure surface the aircraft flies on, the GNSS difference how high that surface lies: D-values, height contours and the geostrophic wind, and from every climb an independent layer temperature.

4 SELF-CORRECTION & CALIBRATION — learnt from the data itself, checked against independent references

Aircraft hundreds of airframes a day, each with its own small instrument errors ⇄ **Each other** every airframe is compared with the other aircraft of the same layer and hour ⇄ **Jokioinen radiosonde** FMI open data, soundings at 00, 06 and 12 UTC: the independent truth ⇄ **MEPS model** MET Norway open data, every hour, taken at the aircraft's own positions

Heading offset of each airframe

Many avionics convert their inertial true heading to magnetic heading with a magnetic-variation table that is years old, while the declination at Helsinki has grown from 7.0° (2005) to 10.5° (2026). A heading offset shows as a wind error across the heading, proportional to the airspeed, so each airframe's offset δ is learnt from its wind against the other aircraft of its layer (fleet mean +1.5°, E190 3.5°, B777 ≈ 0°), faded with a 30-day half-life; by type or fleet median until the airframe has two flights.

Wind against the soundings 11.5 → 9.1 kt RMS; single-aircraft scatter halved.

Upper-level temperature

The fleet's Mach/TAS temperature reads about +0.7 K warm at 300–250 hPa — against both the model and the soundings, steady for a month. A correction per level (400–200 hPa) is learnt every hour from the MEPS check of the last 21 days and added when the Skew-T and time–height are served, faded out below FL180. A per-airframe temperature calibration was tested and rejected: it gained only 1 %.

200–300 hPa error against the soundings 1.27 → 0.96 K RMS.

GNSS heights of the pressure surfaces

After every sounding the aircraft that passed a level within ±1 h and 150 km of the drifting balloon are compared with its measured height of that surface. The fit gives a curve per 5 000 ft band — the altimeter's speed-dependent static-pressure error — and a small, shrunk offset per airframe, refitted from 30 days. The EGM96 geoid (about 18 m here, sloping 3 m per 100 km) is removed at every aircraft position.

Within ±2 m of MEPS at 400–250 hPa, hourly correlation 0.996–0.999.

Airspeed, declination and the checks

When an aircraft sends Mach but no TAS, the airspeed uses the live ISA deviation measured by the other aircraft per 5 000 ft band. Every sounding gets a **Sonde check** and every hour a **MEPS check**, with the model taken where the aircraft actually flew. Raw values are always stored and the corrections are applied when served, so they can be learnt again and checked independently — and everything works without the model.

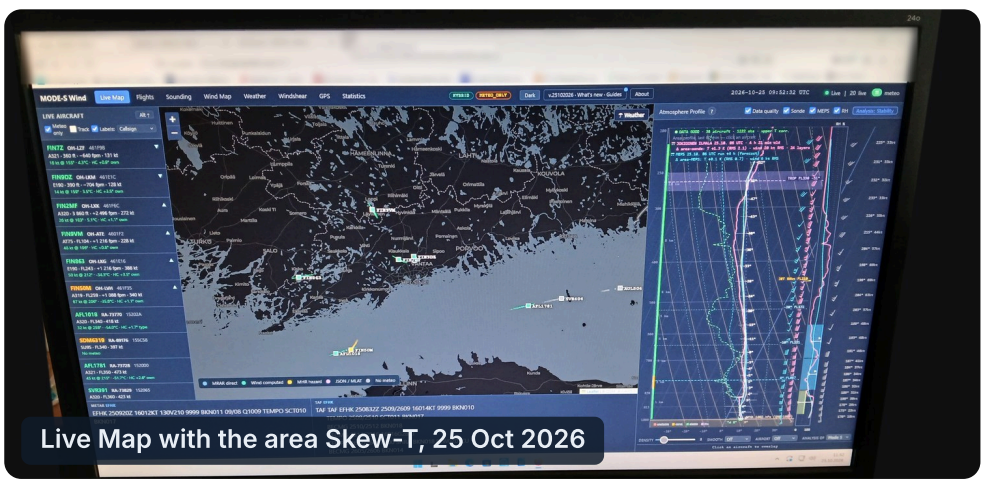
Area wind within 2.4–3.1 kt (vector RMS) of MEPS at 925–400 hPa.

5 DATA STORAGE — SQLite (WAL) on each server

Table / data	Content	Kept
observations · flights	per-aircraft rows: wind and temperature (raw and corrected), GNSS difference, sources	configurable
profile_hours	hourly area profile archive (Skew-T, time–height)	indefinitely
approach_history	every landing: wind per 200 ft, ACI, windshear report	indefinitely
gnss_hours · height_series	hourly height cells · dense climb / descent series	long · 30 d
heading_cal · t_correction · gnss_cal	the learnt calibrations, saved hourly	current
meps_* · soundings/	model profiles (kept) and maps · radiosondes as files	maps 7 d · 1 year
gps_quality_* · gps_episodes	GNSS degradation by hour and episode	manual purge
msg_stats_hours	receiver statistics, one row an hour	7 d

Light on the SD card by design. The live aircraft state, the windshear and GPS trackers and the message statistics live in RAM; a per-aircraft write throttle and an optional meteo-only mode limit the rows; one batched writer thread and hourly jobs that write one row per hour. About 12 MB of database a day; the Maintenance page shows every table and purges by date.

6 WHAT THE SYSTEM SHOWS



Live Map with the area Skew-T, 25 Oct 2026

Live Map aircraft by meteo source, wind barbs, MEPS or radar weather, receiver coverage, area Skew-T; ATC style — page 7

Sounding Skew-T now, any hour or per flight; time–height (also the boundary layer); indices; sonde and MEPS checks

Wind Map wind and temperature at a level, isotachs, streamlines, heights, geostrophic wind, cross-sections

Windshear ILS profile, headwind-ramp alerts (ADV / WSA / MBA), go-arounds, ACI — page 3; EDR, boundary layer — pages 6, 8

Weather · GPS MEPS maps 48 h · GFS 16 days and meteograms (viewing) · GNSS jamming monitor (Full view)

Statistics · /app formats, registers, interrogators, range rings, system health · the phone view — page 5

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7 WINDSHEAR DETECTION — from the aircraft's own replies, for every landing and departure

Low-level windshear is a sudden change of the wind along the approach or climb-out path. ICAO counts it below 1 600 ft above the runway; a headwind change of 15 kt or more, lasting longer than a gust, is significant. What matters to an aircraft is the change of the **headwind along its own path** — and that is exactly what each aircraft reports in its **BDS 5,0** replies: the difference of its true airspeed and ground speed.

The station follows this headwind about every two seconds along the path of every aircraft on final and on the climb-out (noise about 1.1 kt per reply, a new value about every 290 m). The method follows the **Hong Kong Observatory**, which finds headwind **ramps** along the glide path in Doppler-lidar scans — the lidar's one-dimensional headwind profile is the same thing an approaching aircraft samples itself.

`headwind = TAS − GS (BDS 5,0 reply; v8: ADS-B GS — page 8)`

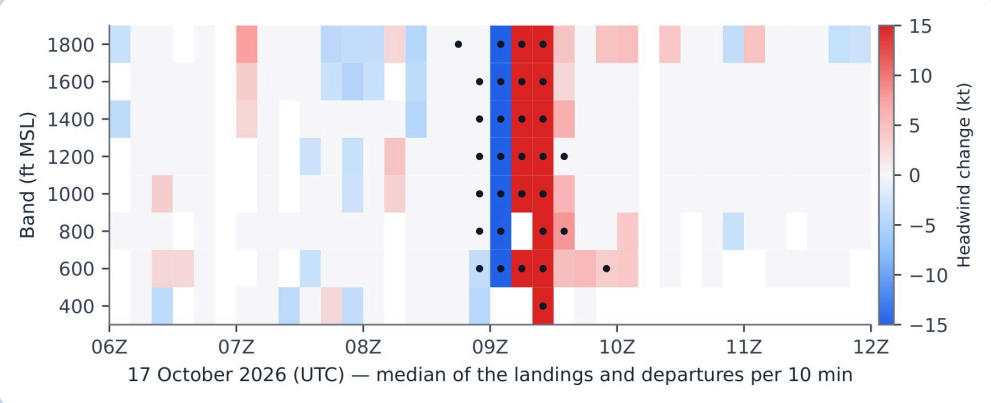
$\Delta V_{1\ km} = \Delta V \cdot (1\ km\ /\ L)^{1/3}, \quad L = 400 \ldots 4\,000\ m$

Replies in a bank, repeated registers and spikes are dropped; the profile is a 400 m running mean along the path. The strongest loss and gain of each flight are scaled to a 1-km equivalent, so a quick change counts more than a slow one. Only the zone below 1 600 ft above the threshold or within 5 NM counts.

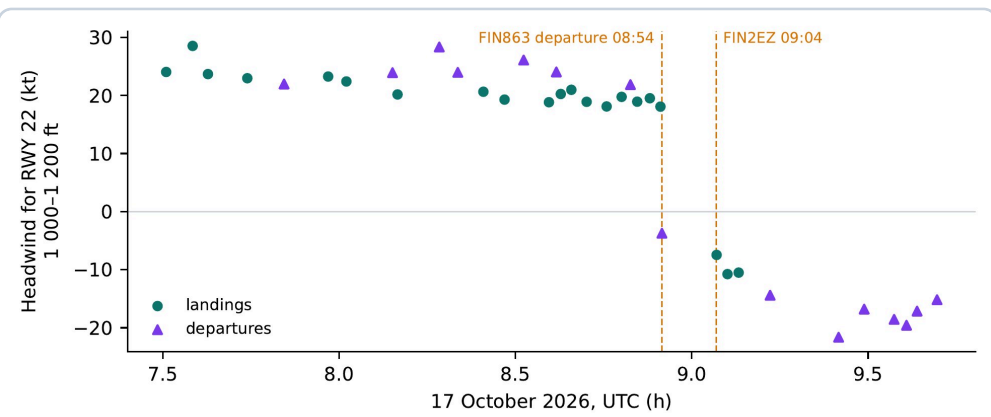
ΔH_{band} = headwind(200-ft band) − median of the flights of the last 45 min

A front or gust front shows as a change *between* flights: each 200-ft band from 400 to 1 800 ft is compared, on the wind vector, with the landings and departures of the previous 45 minutes — landings with landings and departures with departures, because the airspeed reads about 1.4 kt differently in the two flight phases.

Weak ramps (10–12 kt) count only when they stand out from what the other flights met at the same heights; each ramp alert is marked **confirmed** when another landing within 15 minutes saw the same change at the same height, or **isolated** (probably turbulence) when none did. The detection runs on the Pi for every flight, whether or not a page is open; every landing and departure gets a stored **windshear report**, and the Windshear page and `/app` show live alerts, a 24-hour log and the ILS profile.



The front of 17 Oct 2026, band by band (space–time view): each band's headwind change against the previous 45 min — at 09 UTC a blue streak through every band, then the red reversal after the runway change; dot: at or beyond the advisory limit.



A change between flights, 17 Oct 2026. Headwind at 1 000–1 200 ft of the landings and departures on runway 22: a departure met 25 kt less headwind ten minutes before the first landing; three landings in a row then confirmed it (band WSA), before the wind reversed and runway 04 came into use.

ADV advisory ≥ 10 kt **WSA** alert ≥ 15 kt (ΔV ≥ 14) **MBA** ≥ 30 kt loss

The alert message — in the short form that airport windshear systems (LLWAS, TDWR, Hong Kong) give to the tower:

`15A WSA 15K+ 2MF` *"Runway one five arrival, windshear alert, one five knot gain, two mile final"*

Runway + **A**rrival / **D**eparture · level · change in kt (**−** loss, **+** gain) · where first met: RWY, n-mile final or, climbing out, the altitude. Marked *research*.

8 THE ONGOING STUDY — windshear from passively received Mode S EHS at Helsinki-Vantaa

No published study has detected low-level windshear from Mode S EHS replies received passively. MODE-S Wind tests it at EFHK, and every step is documented with its reason and checked on the stored flights — a **study log** for aviation meteorologists and atmospheric scientists, updated as the study goes on (`doc/EFHK_MODE_S_EHS_Windshear_Detection_Study.pdf`; how to read the page: `doc/Windshear_User_Guide.pdf`).

May – 24 Oct 2026
Six browser algorithms
3 084 calm landings: they mostly saw the normal wind increase with height — retired.

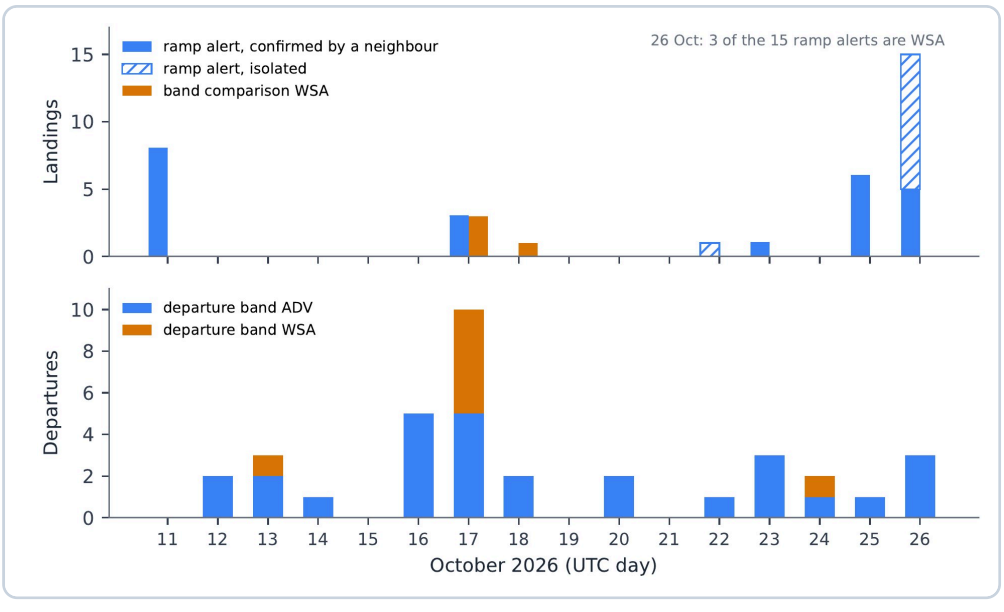
24 Oct
Ramps and band comparison
The Hong Kong method on the server; bands on the wind vector with departures as references.

26 Oct
Weak advisories relative
A stable evening: 10 of 39 landings flagged for the normal profile; 59 → 26.

27 Oct
Confirmed or isolated
A turbulent day: one shear layer seen by four landings among single-aircraft eddies.

27 Oct
By flight phase
Departures read +1.4, landings −1.5 kt against the MEPS model — an airspeed effect, now removed.

28 Oct
14-kt rule · ICAO form
As at Hong Kong, an alert needs a real 14-kt change; messages in the controllers' form.



Windshear flags per day, 11–26 Oct 2026 (current method on all stored flights). Top: landings — ramp alerts confirmed or isolated, and band-comparison WSAs. Bottom: departures.

3 778 · 3 539
landings · departures with a windshear report, 9–27 Oct 2026

36 · 25 ✔
landings with a ramp alert · of them confirmed by a neighbour (11 isolated)

0.9 % · 0.03 %
landings with an advisory · with a ramp windshear alert (14-kt rule)

232–733 ft
lowest height followed above the threshold, by runway

Next

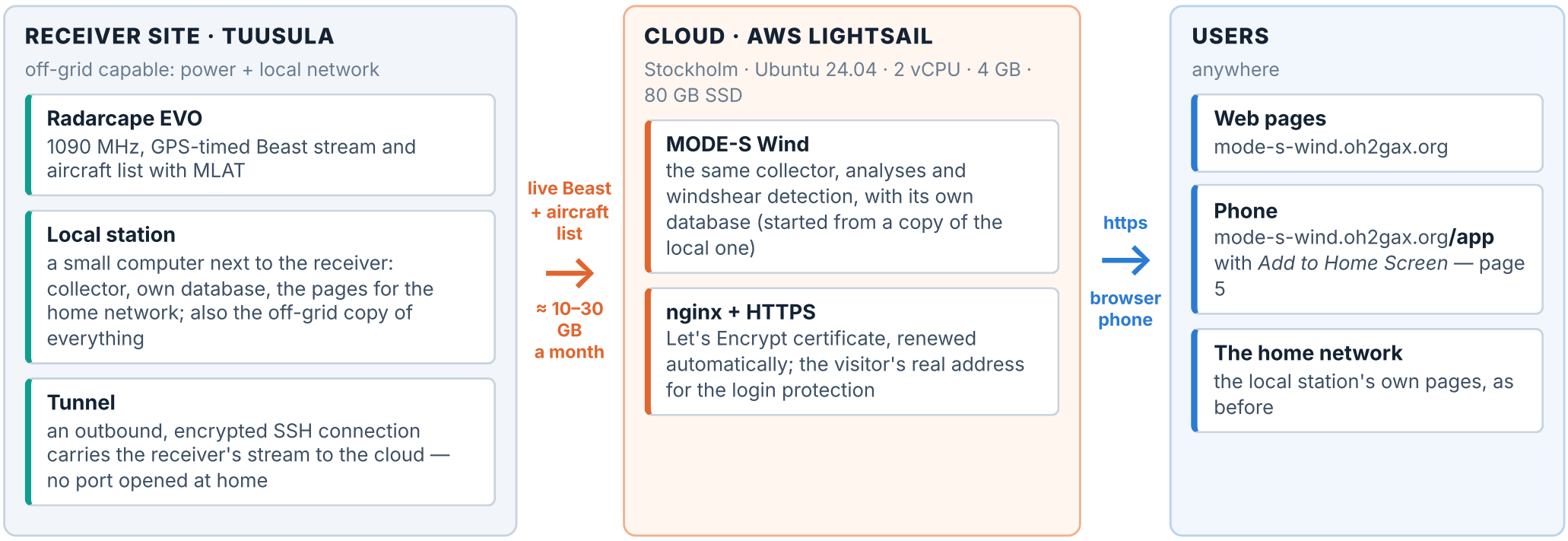
- **ADS-B ground speed** (about twice a second) to sharpen the ramps — **in use from 31 Oct 2026** (detection v8) — page 8
- **Convective season 2027**: gust fronts and microburst outflows are the real test of the levels
- **Validation** against the METAR windshear groups and the pilots' reports (stored from 29 Oct 2026) and go-arounds; turbulence and the boundary layer — page 6

Research only — passive and uncertified; the alerts describe air already flown through. Not for flight operations or ATC.

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9 ONE SYSTEM, TWO SITES — from 27 October 2026

MODE-S Wind now runs in two places with the **same software**. The **local station** at the receiver site keeps the original idea: with power and the local network alone, the receiver's data flows into its own database — no internet needed (only the model, sounding and METAR downloads wait for it). A **cloud server** at Amazon Web Services runs a complete second copy of MODE-S Wind, fed live from the same receiver, and serves all users at **https://mode-s-wind.oh2gax.org**.



One code base on GitHub (oh2gax/mode_s_wind) — both servers run the same version; a change is tried at the local station first and reaches the cloud when it runs well. The few settings in which the servers differ are kept on each server.

	Local station	Cloud server
Role	development station; off-grid observing site	production: the public service for all users
Network	home fibre; the home network	data-centre network, 4 TB of traffic a month included
Without internet	keeps collecting, analysing and serving the home network	no new data until the tunnel is back
Database	SQLite on the station; the reference of the data checks and the windshear study	SQLite on SSD; backed up before every update
Calibrations	learnt independently on each server (heading offsets, upper-level temperature, GNSS heights) — they stay close, and each can be checked against the other	

2 copies of the database, each complete — at the receiver site and in the cloud

1 tunnel outbound and encrypted: the cloud sees the receiver as if it were next to it

≈ 10–30 GB of Beast data a month to the cloud, plus the aircraft list every 2 s

https a permanent, secure address with a certificate — also for the phone view

10 WHY A CLOUD SERVER

- **Room for many users.** Every open page asks for new data every few seconds and the Live Map keeps a live stream open. A Raspberry Pi on a home connection serves a handful of users; the cloud server has a data-centre connection and its own processor, memory and SSD for them.
- **The receiver site is not exposed.** Users connect to the cloud, never to the home network; the site's only connection is outbound.
- **Secure and simple to reach.** A permanent https address with a certificate, also for the phone view.
- **Two independent copies.** The local station collects even when the internet or the cloud is down; the cloud keeps serving when the station is serviced or replaced.

11 KEPT IN STEP

new commits › database backup › update (fast-forward only) › compile check › restart › log check

Every update goes the same way: it lists what is new, copies the database, takes the new version from GitHub, checks the changed program files — going back at once if one fails — restarts and checks the log; the previous version can be restored with one step. Settings that differ between the servers live only on each server, so an update never overwrites them.

Development at home, production in the cloud. A change runs first on the local station, with the same live data, before it reaches the users. The local station's former users are pointed to the new address by a notice on its pages.

12 NEXT — growing from one receiver into a network

soon — **Independent tunnel**
the tunnel gets its own small device at the site, so the cloud's feed no longer depends on the local server.

planned — **Filling gaps**
after an outage, each server copies the missing hours from the other — two databases, nothing lost.

2 Nov 2026 — **Second receiver**
an AirSpy in Vantaa, merged into the weather in the cloud: more of the low approach replies — page 9.

later — **Own multilateration**
three or more GPS-timed receivers: aircraft positions independent of GNSS jamming, also at low level.

Status, 27 October 2026. The cloud server went live with HTTPS on its first day, fed through the tunnel from Tuusula and started from the local station's database; both servers were updated from GitHub the same evening, and the users of the local station are being guided to **https://mode-s-wind.oh2gax.org**.

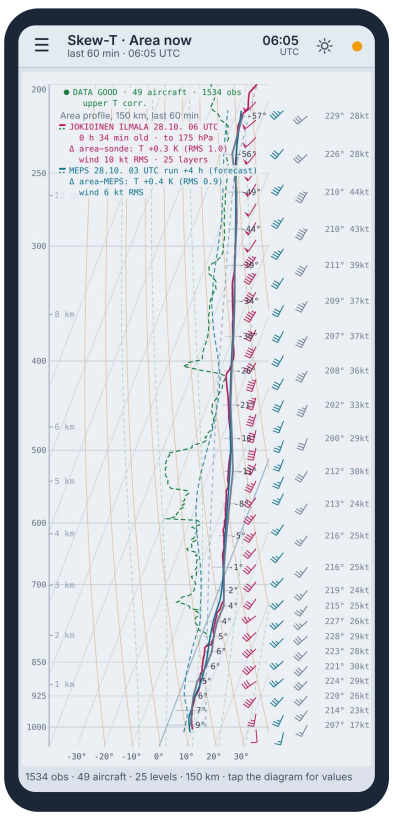
© 2026 Otso Laakso (OH2GAX) · MODE-S Wind, GPLv3 · github.com/oh2gax/mode_s_wind

13 MODE-S WIND ON YOUR PHONE — mode-s-wind.oh2gax.org/app, for all users

The web pages are made for a large screen. For the phone there is a light version of its own, **/app**: open the address of MODE-S Wind with **/app** at the end and log in with the same personal user name and password as on the web pages. It shows **one product at a time, filling the screen**, and every choice is in the ☰ menu — tiles for the views, tap buttons and switches for their options, remembered on the phone. The diagrams are the same as on the full site and come from the same live data: the profiles measured by the aircraft over the last minutes, the approaches and the windshear reports of every landing and departure, and the receiver's own statistics.

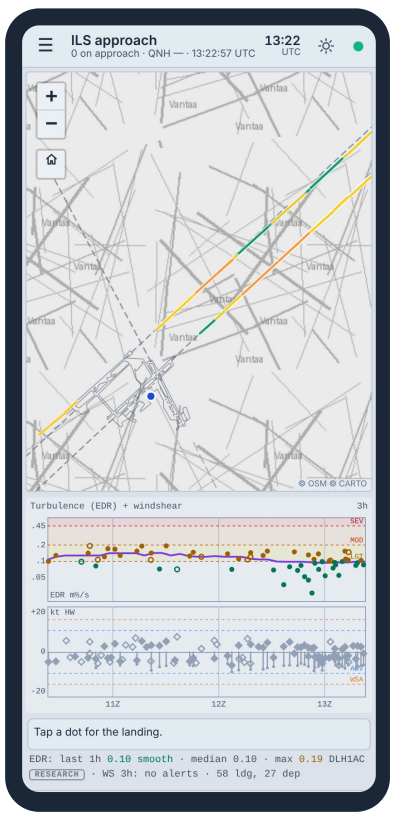
Add to Home Screen (Safari: Share › *Add to Home Screen*; Chrome: menu › *Add to Home screen*) puts a MODE-S Wind icon on the phone that opens the view full screen, like an app — no app store needed. The user guide *MODE-S Wind on a phone* explains every view; the menu's *General* section opens it, the other guides and the *About* text.

mode-s-wind.oh2gax.org/app
scan with the phone's camera



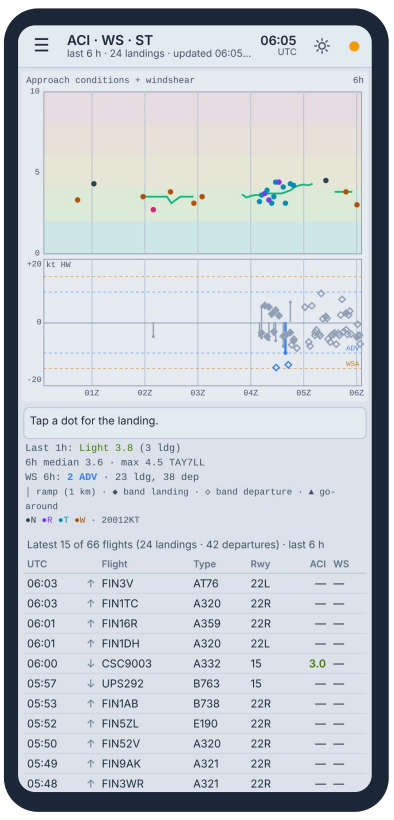
Skew-T · Area now

The temperature and wind measured by 49 aircraft within 150 km over the last 60 min (grey), with the Jokioinen 06 UTC sounding (pink) and the MEPS model (teal): on average the aircraft are 0.3 K warmer, the wind differs by 6–10 kt RMS. Tap the diagram for the values of a level. 28 Oct 2026, 06:05 UTC.



ILS approach · turbulence

The approach map with the last hour's turbulence (research): each 2-km piece of every landing's and departure's path coloured by its EDR — green smooth, yellow light, orange moderate; below, the EDR and windshear of the last 3 h. The live profile and strips are in the same view. 28 Oct 2026, 12:22–13:22 UTC replayed: 32 landings, 2 departures, peak 0.26.



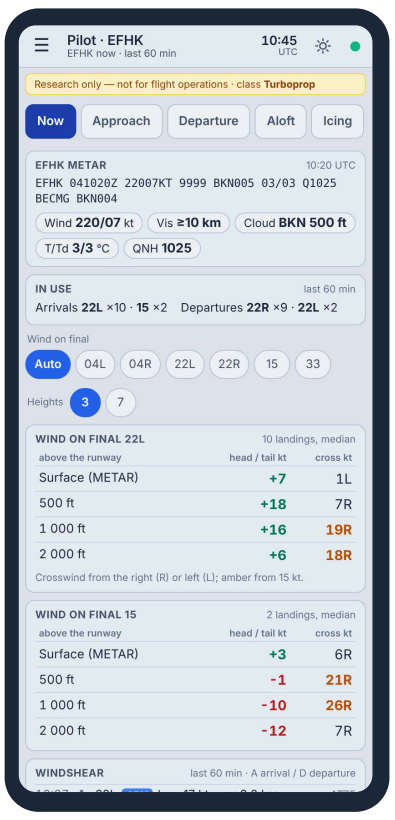
EDR · WS · ACI · ST — ACI + windshear

Top: how rough each landing was (Approach Conditions Index); bottom: the windshear report of every landing and departure as a change of headwind in kt, with the advisory and alert limits. Under it the flights of the period. 6 h to 28 Oct 2026 06:05 UTC: departures climbing into a night-time low-level jet, two advisories (ADV).



Statistics

What the receiver hears: the message rate of the last 24 h, the downlink formats (ADS-B, the Comm-B replies that carry the weather registers, all-call and ACAS) and the range rings; also signal strength, registers, interrogators and the system health.



Pilot · Now (Pilot right)

For pilots, research only: the METAR, the runways in use and the wind on final — head and cross at the surface and 500 / 1 000 / 2 000 ft above the runway, from the last hour's landings; the runway chosen by hand if needed. 4 Nov 2026, 10:45 UTC: a front, 22L and 15.

14 TWELVE VIEWS — one tile each in the menu

Skew-T · now
the area profile of the last 30 min – 3 h, with sonde and MEPS

Skew-T · hour
any stored hourly profile, day and hour steppers

Time–height
hourly profiles side by side, 12 h – 14 days: fronts, inversions, jets; flight by flight

ILS approach
glideslope profile, barbs, alerts, strips; the approach map with turbulence

EDR · WS · ACI · ST
turbulence, windshear, conditions, space–time, boundary layer (also per flight)

History
landings and departures as cards with their winds, headwind profile and turbulence; search

Statistics
message rate, formats, meteo yield, range rings and more

Weather
MEPS maps 6 h back – 48 h ahead, 2.5 km near the airport; the FMI radar; GFS to 16 days

Mast
the Kivenlahti mast's lowest 300 m beside the aircraft, the model and the sounding

Meteogram
one place over the days — EFHK, a receiver site or a tapped place; MEPS and GFS

GNSS
degraded GPS by height, its 7-day history and the spoofing indicators (Full view)

Pilot
EFHK now, approach, departure, aloft, icing, turbulence, my flight (Pilot right)

15 MADE FOR ONE HAND

- **Tap for values** — what the mouse shows on the full site: a level of the Skew-T, a landing on the ACI chart, a cell of the space–time diagram, an aircraft on the approach.
- **Portrait or landscape** — turn the phone for a longer time axis or a wider approach.
- **UTC at the top** — all times in MODE-S Wind are UTC; the clock beside the title shows it.
- **Dark, light or automatic** — one button; *Auto* follows the phone's own setting.

16 LIGHT ON MOBILE DATA

- Only the view on the screen is updated, and nothing while the screen is off.
- The live ILS view loads only the aircraft on the approach, compressed: about **1–4 MB an hour**.
- The landing and departure lists come compressed and without the details the phone does not show.
- The chart program and the map are loaded only when a view first needs them.

17 THE SAME SYSTEM

- The same server and database as the web pages — the phone view is another way to look at them.
- The **health dot** at the top right turns amber or red when a part of the system is late or has stopped; the menu says which.
- Skew-T settings and the theme are shared with the full site in the same browser.

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18 TURBULENCE (EDR) — how rough the air was on every approach and climb-out, from the same headwind replies

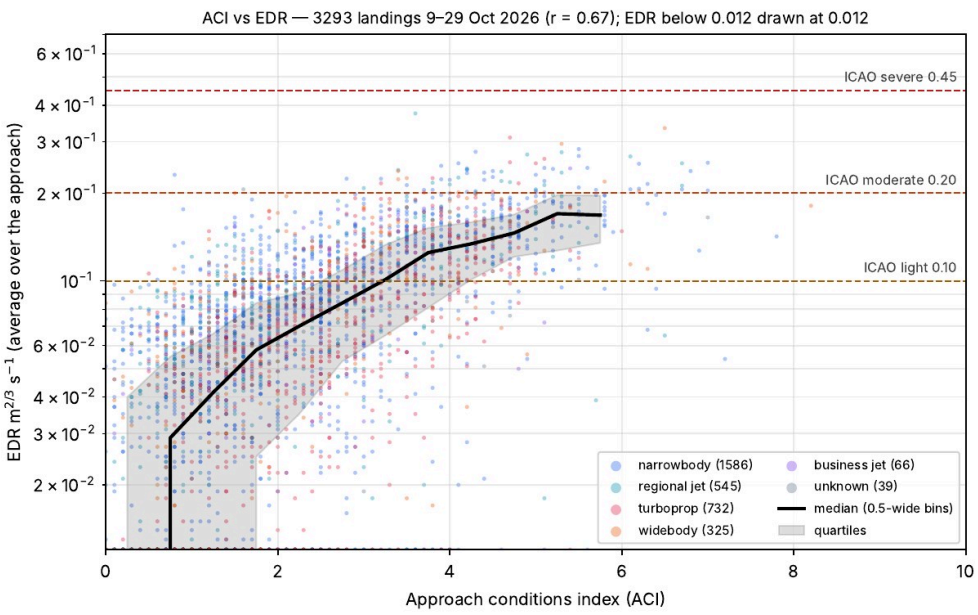
EDR (eddy dissipation rate, $\varepsilon^{1/3}$ in $m^{2/3}\ s^{-1}$) is the measure of turbulence that ICAO uses in the aircraft's own automatic reports: **light** from 0.10, **moderate** from 0.20, **severe** from 0.45. MODE-S Wind computes it for every landing and departure from the headwind series it already follows for the windshear detection — no extra data.

$$\mathrm{EDR} = \sqrt{(D - N_0) / (2 \cdot r^{2/3})},\quad r \approx 141\ \mathrm{m},\quad N_0 = 0.35\ \mathrm{m^2/s^2}$$

D is the mean square difference of the headwind between replies 100–200 m apart along the path, on straight 2-km pieces only (a turn changes the headwind of the steady wind and would read as turbulence). N_0 is the noise of the 2-kt steps of the airspeed and ground speed — the floor of the calmest days. The landing gets the mean of its pieces, a low and a high part and the roughest piece; departures are followed up to 5 000 ft, so most climb-outs get a value too.

It measures **the air, not the aircraft**: the ratio of each aircraft class to its neighbours within 45 minutes is 0.85–1.06 (the approach conditions index ACI needs class factors of 1.0–1.5). It grows with the surface wind — median 0.03 at 0–4 kt, 0.16 at 15–19 kt — and is higher near the ground (0.09 below 1 000 ft, 0.06 above).

Where: the Windshear page's Approach History has an **EDR** view (with a filter for light / moderate), the ACI chart an **ACI + EDR + WS** view, and on the phone the History shows an EDR chip on each card and the approach map colours every landing's and departure's path by its EDR (page 5).



ACI and EDR of 3 293 landings, 9–29 Oct 2026 (r = 0.67; the daily medians agree at 0.98). The black line is the median EDR at each ACI: light turbulence (0.10) is reached at ACI ≈ 3.3.

29 % · 2.4 %
of the landings with light (≥ 0.10) · moderate (≥ 0.20) turbulence on average over the approach

0.145 · 0.065
median EDR of landings with a confirmed ramp alert · without any windshear level

A research estimate. The replies come about every 2 s (140 m apart) — coarser than the scales the definition is based on, so the values are a *Mode S EDR*, still to be calibrated against the pilots' turbulence reports (ARS / WXREP) that MODE-S Wind now collects.

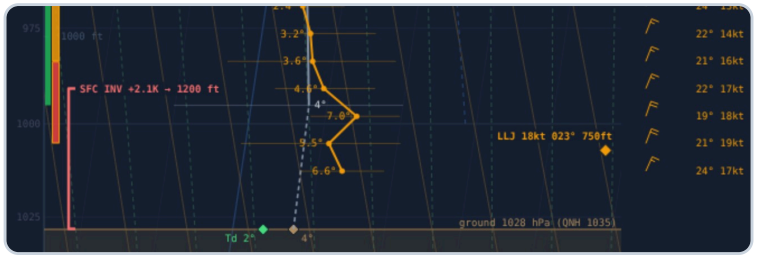
19 THE BOUNDARY LAYER — the low-level jet and the surface inversion, hour by hour

Every hour the station builds the **profile of the lowest 5 000 ft** over the airport from the aircraft themselves: the wind of all aircraft within 25 km in 250-ft layers (the median of at least three replies), and the temperature of the landings and departures in 200-ft layers against the METAR surface temperature. Two features of a stable night are then looked for:

- **Low-level jet** — a band of strong wind a few hundred metres up, mostly at night when the ground cools and the air above decouples from it: a wind maximum at or below 2 000 ft that weakens by at least 2.5 m/s and 25 % within 1 500 ft above it and is at least 4 kt stronger than below (after Baas et al. 2009, made stricter: as such it found a "jet" in 39 % of the hours).
- **Surface inversion** — air above warmer than at the ground: the layer where the temperature rises with height, and its top.

Each hour is stored (about 1–1.5 kB, kept indefinitely). In the month 29 Sep – 29 Oct 2026 (594 hours with traffic) a jet was found in **10 %** of the hours — 16 % at night, 4 % at midday — typically at 1 500 ft and 23 kt; a surface inversion in **15 %**.

Where: the Windshear page's **BL** view; the Sounding page's time–height section (*Boundary layer · wind / temperature* and an overlay on any field); an **LLJ** / **INV** tag on the windshear reports of such an hour; the Skew-T's airport profile (◆ jet, red bracket inversion); the phone's EDR · WS · ACI · ST view; from 31 Oct 2026 **flight by flight** — page 8.



The airport profile on the Skew-T (lowest 2 000 ft): a jet of 18 kt from 023° at 750 ft (◆) and a surface inversion of +2.1 K up to 1 200 ft (red bracket); orange: the temperature of the last hour's landings and departures.

0.5 % · 1.2 % · 0 %
of the landings with a windshear level in jet hours · in other hours · under a surface inversion (2 / 421, 47 / 3 840, 0 / 460) — a stable layer damps the turbulence: median EDR 0.048 in jet hours, 0.019 under an inversion, 0.076 otherwise



Time–height section of the boundary layer, 17–20 Oct 2026 (Sounding page): the wind of each hour and 250-ft layer coloured by speed, with barbs; on the evening of 17 Oct a north-easterly jet of 30 → 24 kt at 1 500–1 800 ft (◆), on the following nights surface inversions (red bars, the line their top). Hatched: no flights.

20 THE ATC-STYLE MAP — the Live Map and the Windshear map as a radar display, from 30 October 2026 (research)

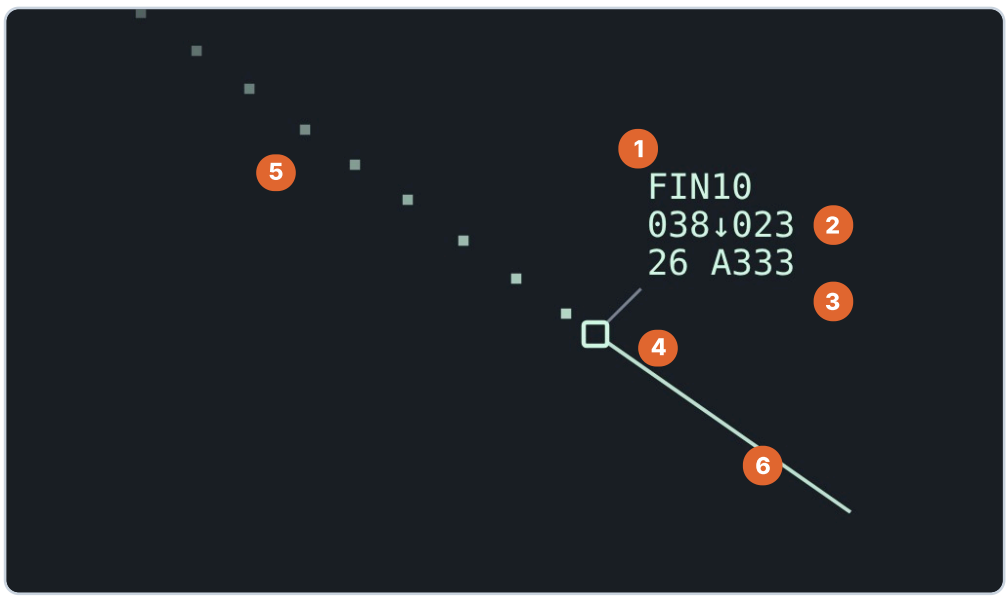
Many of the users have an air traffic control background. For them both maps have an **ATC style** — on the Live Map the map button (*Street* / *ATC map*), on the Windshear page the **style** button, which steps *ATC* (the default) → *Dark* → *Grey*. From 2 Nov 2026 the **labels** are chosen apart, so the ATC labels work on the street map too. The street map gives way to a plain grey background (dark grey in the Dark theme, a very light grey in the Light theme) with only the airport layout, the ILS finals (short dashes), the RNP finals (long dashes) and the coastline; range rings every 5 or 10 NM can be added (**Rings**).

The aircraft are **open squares** as on a radar display — light below 6 000 ft, grey above; a selected aircraft keeps its colour with a frame around its square — with **history plots** behind them (**Hist**: 5, 10 or 20 plots, one about every 6 s), a **speed vector** ahead (**Vec**: 30 s, 1 or 2 min at the ground speed along the track) and a **label** with a leader line. How much the label shows is chosen with **Lbl**: the callsign, registration, ICAO24 address or type alone, two lines (callsign / level) or the full three lines below.

Labels kept apart. Each label tries sixteen positions around its square (eight directions, near and far) and takes the one that overlaps least with the labels already placed, the other squares, the plots and vectors and the panels over the map; it keeps its place as long as that stays free, so the labels do not jump from one update to the next. The selected aircraft is placed first, then the others from the lowest up, so the approach traffic gets the clearest places.

The level is written as on a controller's label, in hundreds of feet: up to the transition altitude of 5 000 ft the altitude on the latest METAR QNH (**038** = 3 800 ft), above it the flight level (**063** = FL063), with ↑ / ↓ when climbing or descending at 300 ft/min or more.

The selected level is the altitude the crew has set on the autopilot panel (MCP / FCU), downlinked in the Mode S register BDS 4,0 when the radars ask for it, or in the ADS-B target-state message; it is shown when heard within the last 2 minutes. It is *not* the cleared level of the controllers' system, but it follows the clearance closely: in a capture at 11:32 UTC on 30 Oct FIN122's selected level changed from 050 to 030 in flight. On final the selected level is often above the aircraft — many crews set the missed-approach altitude once established (FIN8SM in the map below: **018↓020**).

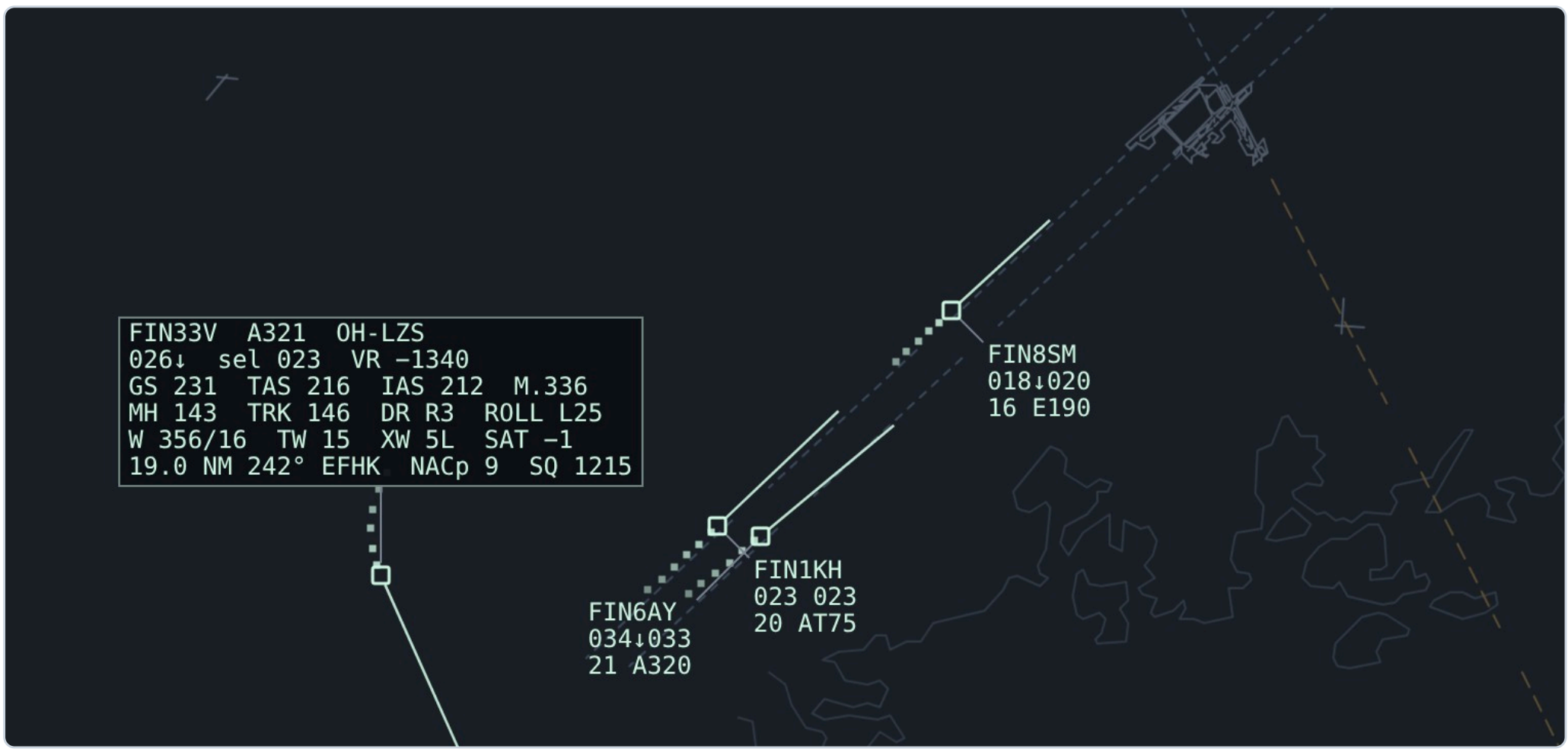


Reading the label (Lbl Full) — FIN10, an A330-300 descending towards Helsinki-Vantaa:

1. **FIN10** — the callsign (ICAO24 address when none is known)
2. **038↓023** — at 3 800 ft on the QNH, descending; 2 300 ft selected on the autopilot panel
3. **26 A333** — ground speed in tens of knots (260 kt) and the ICAO aircraft type
4. the open square at the latest position, the leader line to its label
5. history plots, one about every 6 s, older ones fainter (Hist 10)
6. the speed vector: where the aircraft is in 30 s at its ground speed (Vec 30 s)

Hover — the extended label (from 30 Oct 2026). With the **Hover** button on, the label of the aircraft under the mouse grows into six lines — and from 31 Oct always that of the selected aircraft, in grey — as a controller's extended label — what the station has just heard from it, refreshed every 2 s (FIN33V in the map below):

1	callsign · type · registration
2	level · selected level · vertical rate · AP / APP
3	ground speed · true and indicated airspeed · Mach
4	magnetic heading and track · drift · bank (L25 = 25° left)
5	wind · tail- or headwind · crosswind · temperature
6	distance and bearing from EFHK · NACp · squawk



Live traffic with Hover, 30 Oct 2026, 13:23–13:24 UTC, as received by the station, in the ATC style with Lbl Full, Hist 10 and Vec 1 min (QNH 1019). The mouse is on **FIN33V**, an A321 turning left at 25° of bank (**ROLL L25**), descending through 2 600 ft towards its selected 2 300 ft with a 15-kt tailwind on this track (**TW 15**: ground speed 231 kt, true airspeed 216 kt). FIN6AY and FIN1KH on the finals of 04R and 04L, FIN8SM ahead on 04L with 2 000 ft set — pointed at, its label reads **APP**: the autopilot in approach mode. Stream and hover replies captured from the station, redrawn on a test copy; four aircraft without a position update in the last minute left out.

> **A research display, not for air traffic control.** The positions are what this one station receives — ADS-B and the receiver's MLAT, updated every 3 s — without the radar data processing, flight plans, cleared levels or the safety nets of an ATC system; a gap in reception is a gap on the map. Do not use it for separation or any other operational decision. The style follows the conventions of European radar displays such as Thales TopSky, whose screens and settings are not public — it is not a copy of any one system. Comments from controllers on what to show and how are welcome.

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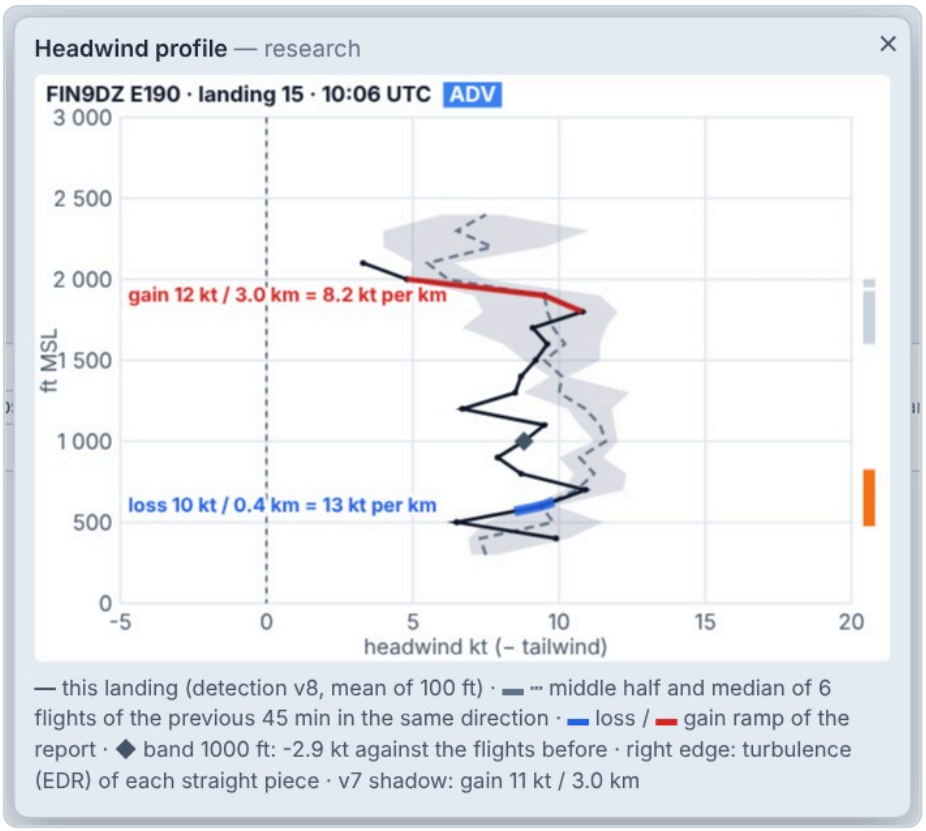
21 WINDSHEAR DETECTION V8 — the aircraft's own ADS-B ground speed, in use from 31 October 2026

Up to now each headwind value came from one BDS 5,0 reply: its true airspeed minus its ground speed, about every 2 s. But every aircraft also **broadcasts its ground speed and track by itself**, about twice a second, in the ADS-B airborne velocity message — no interrogation needed. These have been stored with the raw series of every landing since 24 Oct and of every departure since 27 Oct 2026.

> $\text{headwind} = \text{TAS}_{\text{BDS 5,0 or 6,0}} - \text{GS}_{\text{ADS-B}}(0.5\text{ s before the reply})$
>
> The true airspeed of both Comm-B registers: BDS 5,0, and BDS 6,0 with its indicated airspeed turned into TAS (pressure altitude and the flight's own temperature, or that of the flights nearby); the small offset between the two registers of each flight removed. A flight without ADS-B ground speed is analysed as before (v7).

This gives a headwind about **every 0.5 s instead of every 2 s** — 2.3 times the values, with a little less noise — and the short ramps of 400–600 m are measured better. After one day beside the former method it became the detection on 31 Oct 2026 (**v8**); the former result (v7) is still kept with each report, without alerts, for comparison. The ramps, the band comparison, the levels and the 14-kt rule (page 3) apply unchanged.

Message times checked. The same day the receiver began to check every message's time against the clock: 0.1–0.7 % of the replies carried a time 3–50 s ahead, and a climb-out could then read one reply again at several heights as a gust (FIN3A, 30 Oct: EDR 0.34 instead of 0.07).



One landing's headwind profile — FIN9DZ, an E190 on runway 15 at 10:06 UTC on 27 Oct 2026 (ADV): the detection v8 values, mean of 100 ft (black), against the median (dashed) and the middle half (grey) of the 6 flights of the previous 45 min; its ramps drawn thick — a loss of 10 kt in 0.4 km at about 550 ft (13 kt per km, the advisory) and a gain of 12 kt over 3 km — and the turbulence of each piece at the right edge. Opened by a click on the flight in the Approach History, or on the phone in an opened History card.

0.5 s · 2 s
a headwind value about every — with v8 · with v7 (BDS 5,0 alone)

41 · 32 + 1
ramp advisories · v7's advisories + WSA, 1 233 landings 24–31 Oct 2026 (19 flagged by both)

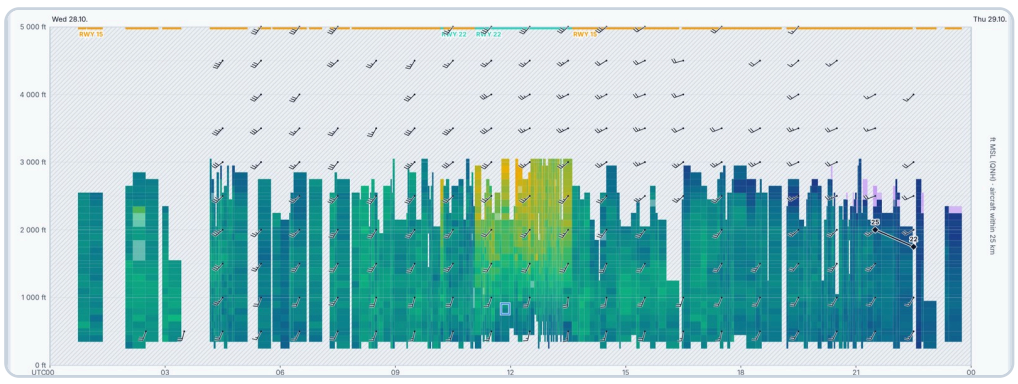
22 THE BOUNDARY LAYER FLIGHT BY FLIGHT — every landing and departure as its own column, from 31 October 2026 (research)

The windshear report of every flight now keeps its **fine headwind profile**: the mean of the headwind every 100 ft from 200 to 5 000 ft (about 200–300 bytes, no extra writes); a departure's over its whole climb-out to 5 000 ft. The hourly boundary layer (page 6) shows the typical air; these show **what changed from one aircraft to the next**. Three fields, each flight a thin column held until the next one (at most 20 min):

- **Headwind** — along the flight's own runway (violet a tailwind); a line along the top gives the runway in use
- **Headwind change** — against the median of the flights of the previous 45 min in the same direction: fronts, a jet forming or dying, a single gust
- **Wind shear** — how fast the flight's headwind changes with height (kt per 100 ft over ±200 ft): the shear layers themselves

Where: the Sounding page's time–height section (*Boundary layer · headwind / headwind change / wind shear per flight*, all flights, landings or departures, up to 14 days); the Windshear page's **BL** view (**HW**, **ΔHW**, **Shear**) beside the low-level jet and the inversion; on the phone in Time–height and in EDR · WS · ACI · ST. Windshear alerts are outlined; hovering — or tapping — a column gives the flight, its headwind at that height, the change and the shear.

Read together they show a layer as the aircraft meet it one after another: on 28 Oct (below) the headwind grew with height in the lowest 1 000 ft through the day (red, the friction layer); in the evening a low-level jet of 25 → 22 kt formed at about 2 000 ft, and the headwind fell off above it (blue) — the top of the jet.



Headwind per flight, landings, 28 Oct 2026 (Sounding page, with the hourly barbs): runway 15 most of the day with 10–20 kt; on runway 22 around midday 25–35 kt (yellow) — the runway, not the wind, changed (top line). In the evening a light tailwind at 2 500 ft (violet) above the jet (◆).



Wind shear per flight, the same day: red where the headwind grows upwards, blue where it falls off — the friction layer by day, the top of the jet at 2 000–2 600 ft from 17 UTC. Pale: within ±0.7 kt per 100 ft.

> **How to read it — research.** A column is what one aircraft measured on its way — about one value per 100 ft on a climb-out, five on an approach — not the whole layer: a single gust is one coloured column, a real layer or front a band that the flights draw one after another, at the same height or slowly rising or sinking. The values are research data from a passive receiver, not for flight operations.

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23 MORE THAN ONE RECEIVER — from 2 November 2026

A receiver hears what its antenna can see. Receivers in different places see different parts of the sky — especially close to the ground, on the last part of the approach — and they back each other up. From 2 November 2026 MODE-S Wind can combine **several receivers into one weather**: each receiver sends its live stream through its **own outbound, encrypted tunnel** to the cloud server — nothing is opened at any site — and the server merges the streams into **one stream of replies**. The winds, temperatures, windshear reports and turbulence are then computed from it exactly as before. The local station at the first site keeps reading its own Radarcape.

MODES-RX-01 · TUUSULA
Jetvision Radarcape — stamps every reply with **GPS time**; the main receiver and the clock of the whole system.

MODES-RX-02 · VANTAA
AirSpy R2 software-defined radio on a **Raspberry Pi 5** — its own clock; on a 4G link — **MODE-S Wind Tactical**, the mobile station (page 10).

MORE RECEIVERS
any receiver that gives the standard Beast stream with a time stamp — a second Radarcape is ready for its site.

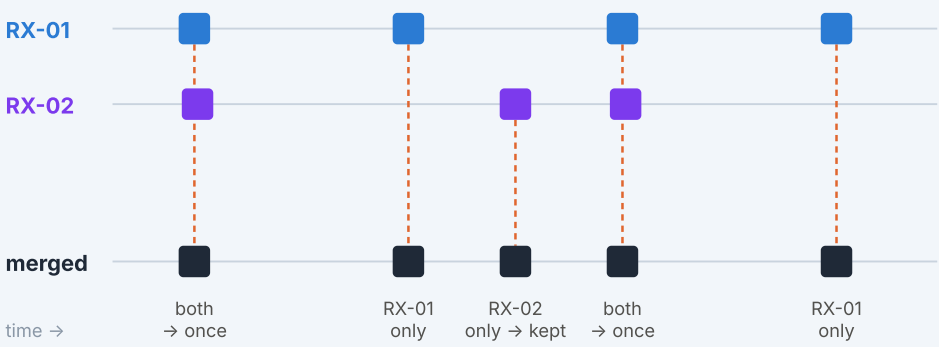
own tunnels → live streams

CLOUD SERVER · MERGER

- **One clock** — every reply dated on the Radarcape's GPS time
- **Each reply once** — a reply heard by several receivers is taken only once
- **In time order** — into the weather, as from one big receiver
- **Statistics per receiver** — what each receiver adds, its clock and its link

24 HOW THE MERGE WORKS

1. **Everyone listens.** An aircraft's reply goes out in all directions, so the same reply often reaches two or more antennas, a fraction of a millisecond apart. Each receiver passes on everything it hears.
2. **One clock.** The Radarcape dates every reply with GPS time. A receiver with its own clock is lined up with it automatically, from the ADS-B messages that both receivers hear: the same message at both is the same moment.
3. **Each reply once.** On that common clock, the same message from two receivers at the same moment is one reply: the copy is dropped. So nothing is counted twice, and a reply that only one receiver heard is kept.
4. **In order, into the weather.** The server waits a moment for the copies coming over the slower link, then passes the replies on in time order. A frame that arrives far too late — after a break of a mobile link — is left out.



Five replies (illustration): the receivers hear them in part, the merged stream has each of them once.

25 WHAT IT GIVES

- **More of the low approach.** Near the ground buildings, trees and terrain hide an aircraft from one antenna but not from another, so with more receivers the replies reach lower on final — where the windshear and turbulence are measured.
- **Fewer gaps.** A direction that is shadowed at one site is covered from another.
- **It keeps going.** When one receiver or its link falls silent, the weather goes on from the others.
- **Any place, any receiver.** A Radarcape, a small software-defined radio, or a mobile station on a 4G link — each only needs its outbound tunnel.

26 WHERE IT SHOWS

- **Statistics** — *All receivers*: what the weather is computed from, each reply once. A receiver's own statistics are shown to its owner (marked *yours*) and to users with full view.
- **Live Map** — the receivers as small triangles and their combined coverage.
- **System health** — a dot for each receiver's feed.

27 NEXT

since 4 Nov 2026
MODE-S Wind Tactical
the AirSpy on a tripod and 4G; off-grid field test 8 Nov (page 10)

ready
Second Radarcape
set up and tested with its own tunnel; to a rural site west of Helsinki, online over Starlink

later
Own multilateration
three GPS-timed receivers: positions independent of GNSS jamming

28 HOW A RECEIVER JOINS

1 · a receiver
Any 1 090 MHz receiver
with the standard Beast output and a time stamp on each reply: a Radarcape, or a software-defined radio with readsb

2 · a tunnel
Outbound and encrypted
opened from the site to the cloud server — by a small computer there, or by the receiver itself; no ports opened, also behind 4G or Starlink

3 · one setting
Added in the cloud
its name, type and clock in the server's list of receivers; its clock is lined up with GPS time by itself within minutes

4 · in use
Visible at once
in the Statistics, the system health and on the Live Map — and its replies go into the weather

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Off-grid in the field · 8 Nov 2026

29 MODE-S WIND TACTICAL — a whole receiving station on a tripod

The second receiver of page 9, **MODES-RX-02**, is built to move. Antenna, receiver, computer and 4G router stand on one tripod; it listens to the airliners overhead — the wind, temperature and altitude they report themselves — and streams the replies live into the merged MODE-S Wind weather. **No fixed network, no public address and no settings on site**: it joins 4G, opens its own encrypted tunnel to the cloud server and starts feeding by itself. At home it stands on a balcony in Vantaa; for a field day it goes into the boot of a car.

3 steps
to deploy — tripod, cable, power

~1 min
from power-on to live data, also after a power cut

~10 W
for the whole station — a day from a portable battery

80–100 MB
of mobile data a day — an ordinary 4G subscription

30 DEPLOY IN THREE STEPS

1. **Mount the tripod.** Stand it up and raise the telescopic mast with the antenna.
2. **Connect the antenna.** One coax cable from the antenna to the enclosure.
3. **Power on.** The 4G router and the Raspberry Pi 5 start; about a minute later the server shows the feed live. Taking it down is the same in reverse.

31 INSIDE

Antenna and cable
1 090 MHz antenna (OPA-5 series, 5 dBi) on a telescopic mast; low-loss H-155 coax.

Computer
Raspberry Pi 5 in a weatherproof enclosure; it keeps the tunnel open and starts again by itself.

Receiver
AirSpy R2 software-defined radio with *readsb* — the standard Beast stream with a time stamp on each reply.

Link and power
Battery-powered mobile 4G router, on line before the station starts; 230 V, or a portable battery in the field.

32 THE FIRST OFF-GRID FIELD TEST — Sunday 8 November 2026, a forest clearing near Helsinki (local time)

11:59	12:31	12:39	12:40	12:46	15:04
Off the balcony	**At the site**	**Connected**	**Live**	**Packed**	**Home again**
switched off and packed into the car with the weekend's shopping	the tripod stood up in a clearing, the antenna low among the trees	the server logged the receiver on line — 8 minutes from arrival, photos included	24 messages a second; in two minutes 2 648 replies from 9 aircraft	switched off at about 12:41, everything back in the boot in 5 minutes	back on the balcony and feeding, the antenna a little higher than before

- **Off-grid power is no problem.** The whole station — receiver, computer and 4G router — drew **9–10 W** from a portable power station's 230 V outlet; at 71 % charge the battery estimated 13.8 hours left.
- **Real data, straight into the weather.** ADS-B and the Comm-B weather registers (BDS 4,0 / 5,0 / 6,0) of the aircraft overhead went into the merged stream like those of any receiver, and more than a fifth of them were heard only there.
- **Proven on live data.** On the balcony it has fed the system since 4 November 2026: 60–95 messages a second by day, about 5 % of all merged replies heard by it alone.
- **Next.** A longer field day on a higher, open site — how far it hears there and what it adds to the low approach — and a direct 12 V supply. The full report: *MODE-S Wind Tactical — first off-grid field test* (user guides).



12:33 · tripod up



12:40 · inside



12:39 · 71 % · output 10 W · 13.8 h left

The field test in pictures. Left: the station up at the site, the enclosure on the tripod. Middle: inside the enclosure — the AirSpy receiver, the Raspberry Pi 5 and the 4G router on one mounting plate. Right: the portable power station a few seconds after the start: 10 W out, 13.8 hours left at 71 %.

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